Final Project: Academy Award Nominated Films, Violence, and the Bechdel Test

2026-04-15

Hypotheses

H1 The higher the level of violence in an Academy Award for Best Picture nominated film, the fewer Bechdel Test criteria the film will pass.

H2 Of Best Picture nominees, winners of the Academy Award for Best Picture are more likely to pass Bechdel Test criteria.

H3 Of Best Picture nominees, winners of the Academy Award for Best Picture are more likely to score a higher level of violence.

H4 The higher the Box Office revenue for an Academy Award for Best Picture nominated film, the more Bechdel Test criteria the film will pass.

H5 The higher the Box Office revenue for an Academy Award for Best Picture nominated film, the film will score a higher level of violence.

Findings

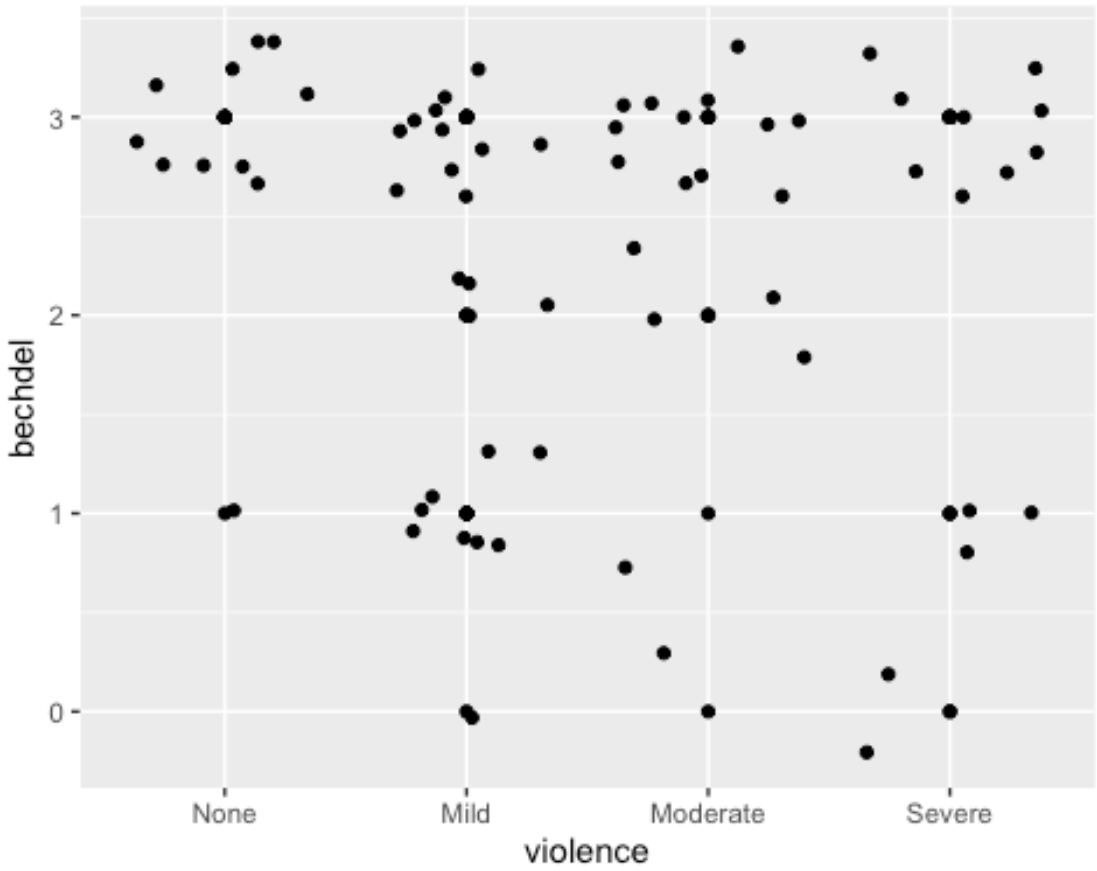
Hypothesis 1

H1 The higher the level of violence in an Academy Award for Best Picture nominated film, the fewer Bechdel Test criteria the film will pass.

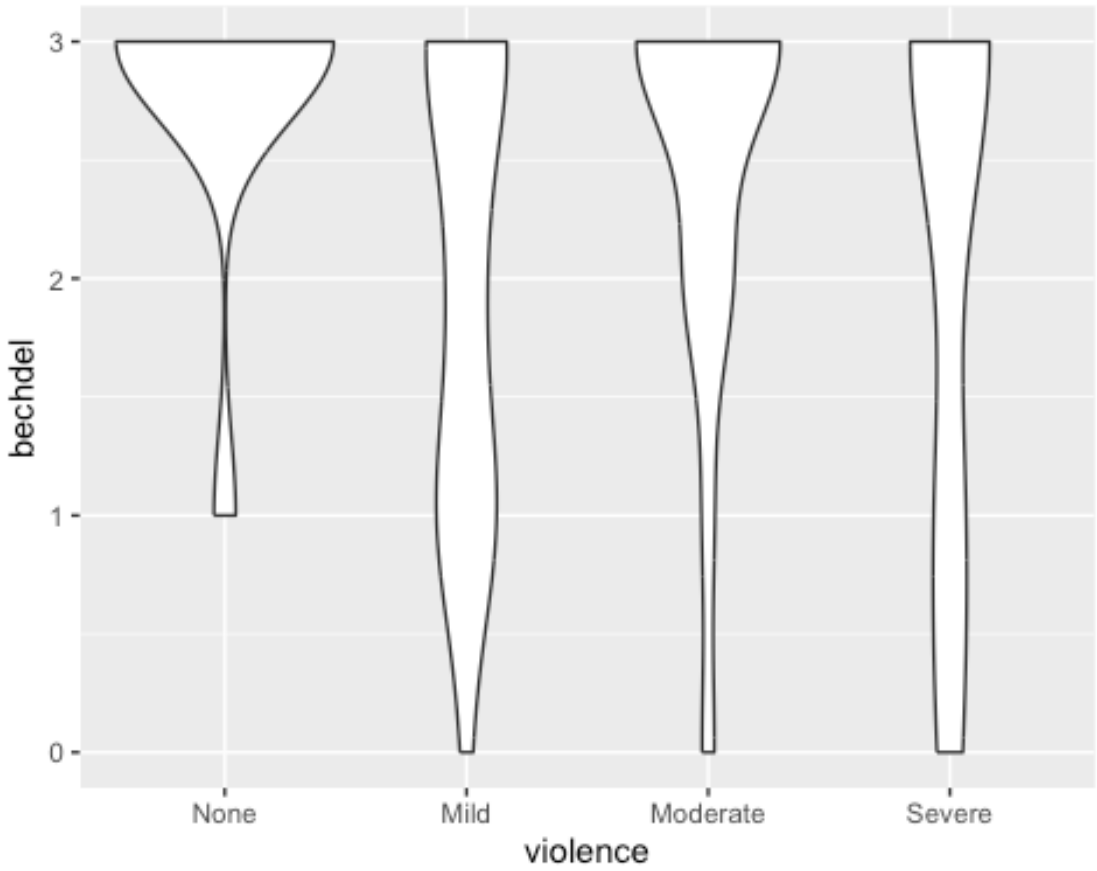
Median and Mean H1

```
## # A tibble: 4 × 5
##   violence     n   Mdn     M    SD
##   <ord>    <int> <dbl> <dbl> <dbl>
## 1 None        11     3  2.82 0.603
## 2 Mild        24     2  2.04 0.999
## 3 Moderate    18     3  2.5  0.857
## 4 Severe      14     3  2.14 1.23
```

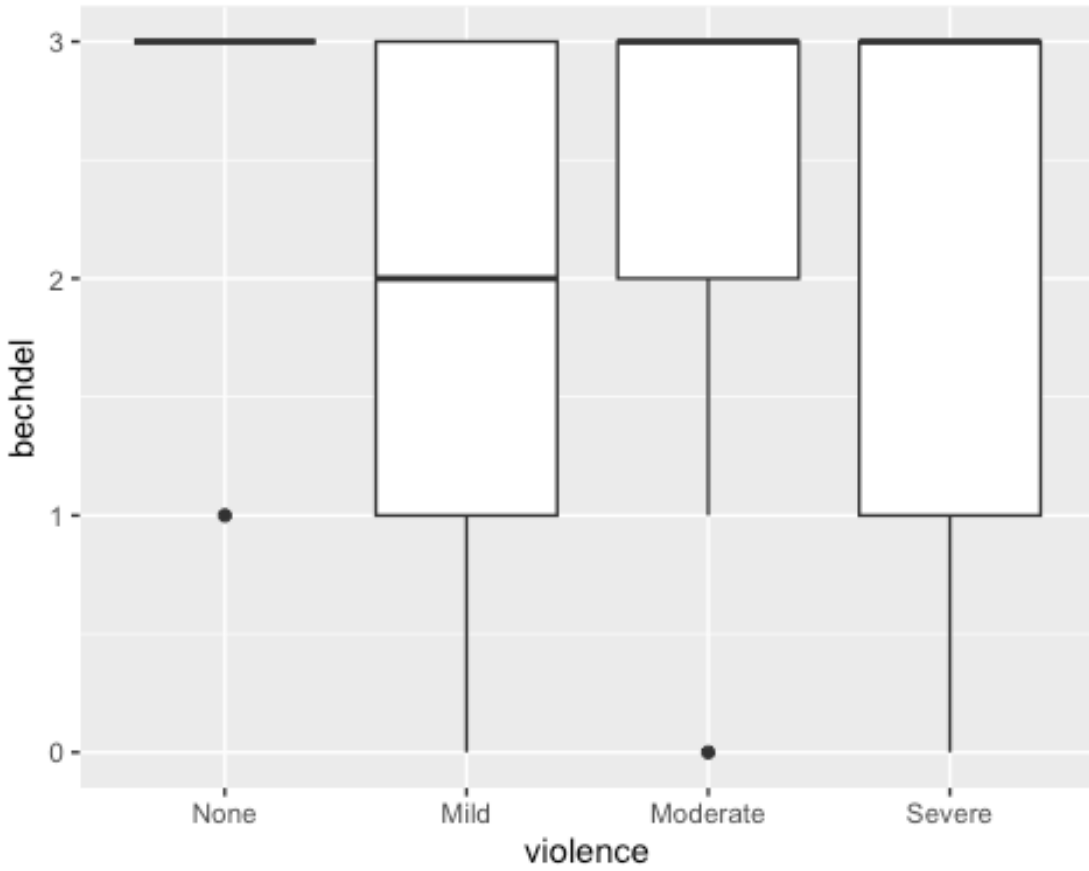
Scatter H1



Violin H1



Boxplot H1



Spearman Test H1

```
## Warning in cor.test.default(x = mf[[1L]], y = mf[[2L]], ...): Cannot
compute
## exact p-value with ties

##
##  Spearman's rank correlation rho
##
## data:  bechdel and violence_coded
## S = 53861, p-value = 0.5478
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##         rho
## -0.07473071
```

Findings H1

There is no real correlation between Bechdel Test criteria passed and violence scores for Academy Award Best Picture nominees, because p = 0.548. PLEASE CHECK.

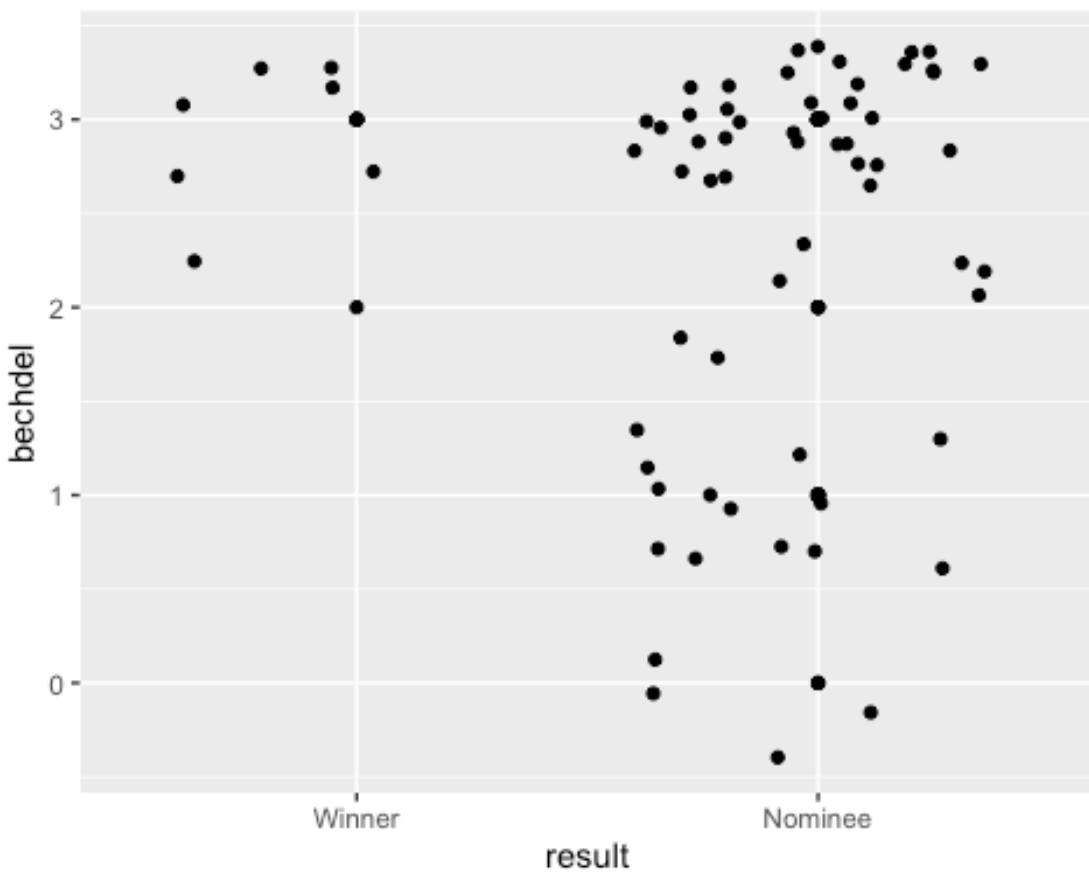
Hypothesis 2

H2 Of Best Picture nominees, winners of the Academy Award for Best Picture are more likely to pass Bechdel Test criteria.

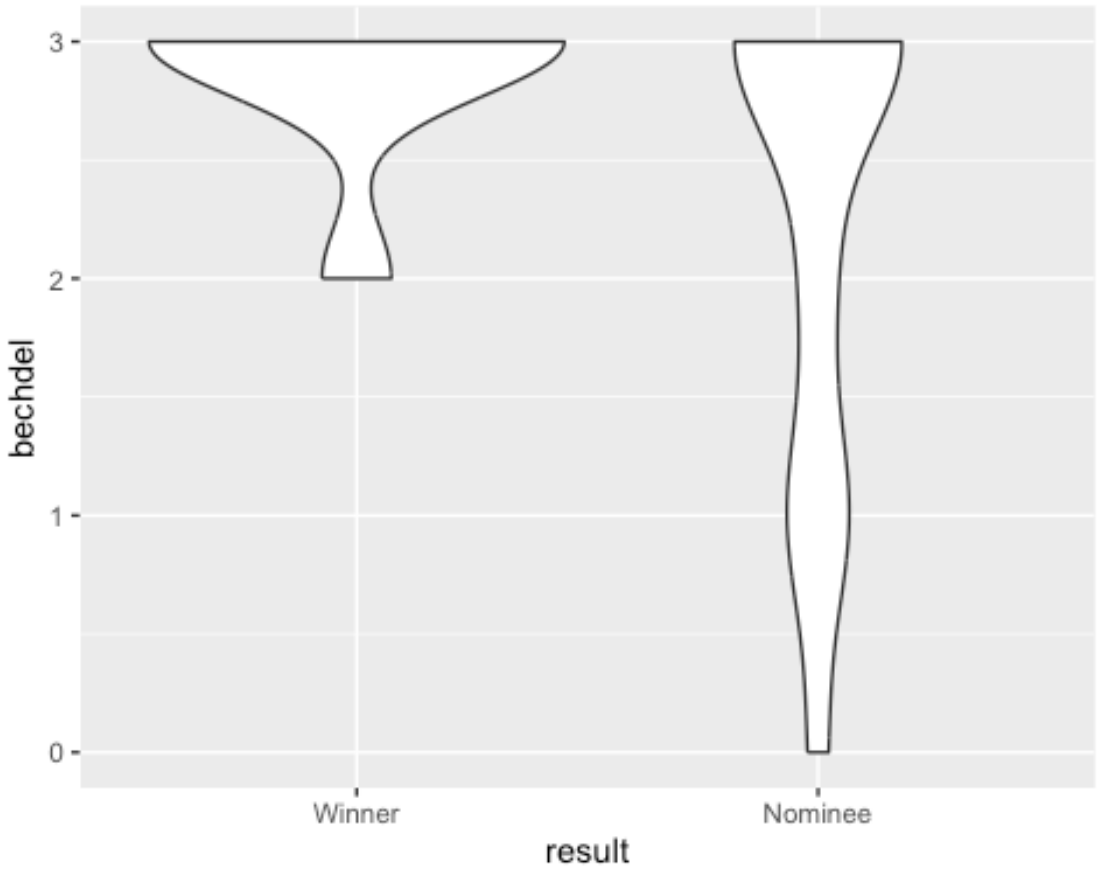
Median and Mean H2

```
## # A tibble: 2 × 5
##   result      n   Mdn     M    SD
##   <ord>   <int> <dbl> <dbl> <dbl>
## 1 Winner      7     3  2.86 0.378
## 2 Nominee    60     3  2.25 1.02
```

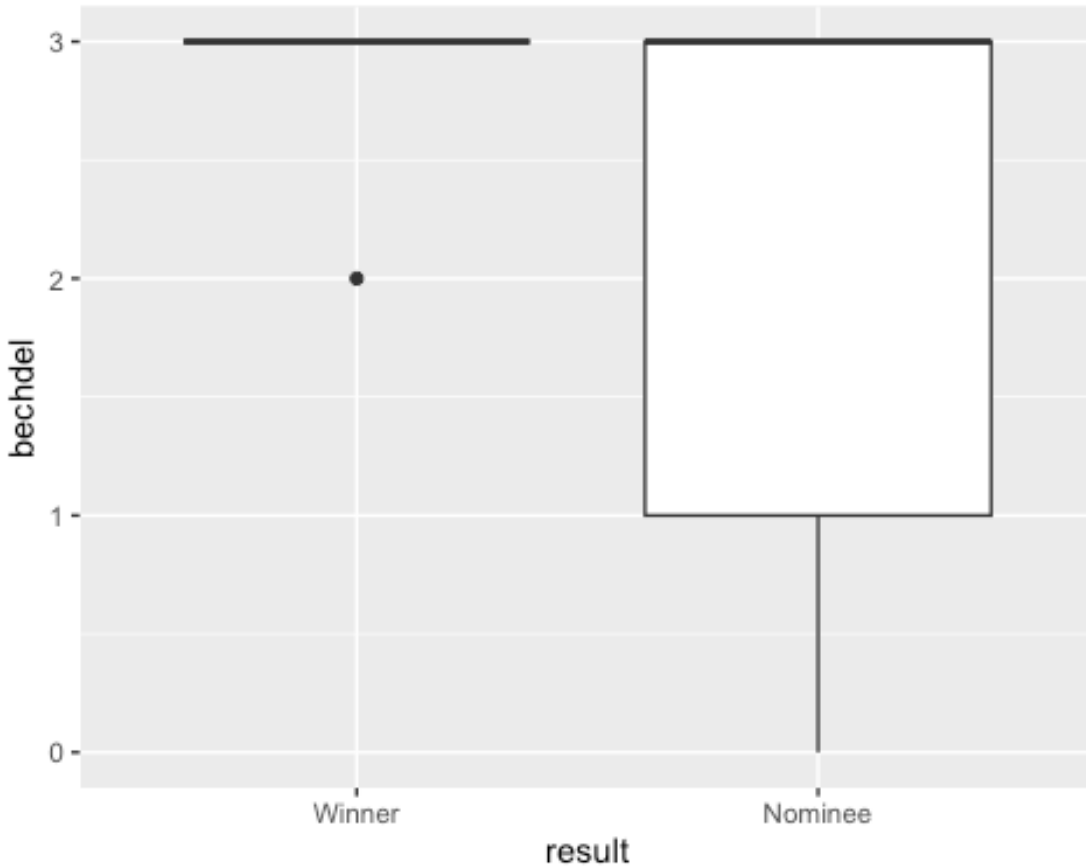
Scatter H2



Violin H2



Boxplot H2



Spearman Test H2

```
## Warning in cor.test.default(x = mf[[1L]], y = mf[[2L]], ...): Cannot
compute
## exact p-value with ties

##
##  Spearman's rank correlation rho
##
## data:  bechdel and result_coded
## S = 59275, p-value = 0.1388
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##        rho
## -0.1827487
```

Findings H2

There is no real correlation between Bechdel Test criteria passed and winning the Academy Award Best Picture among Academy Award Best Picture nominees, because p = 0.139. PLEASE CHECK.

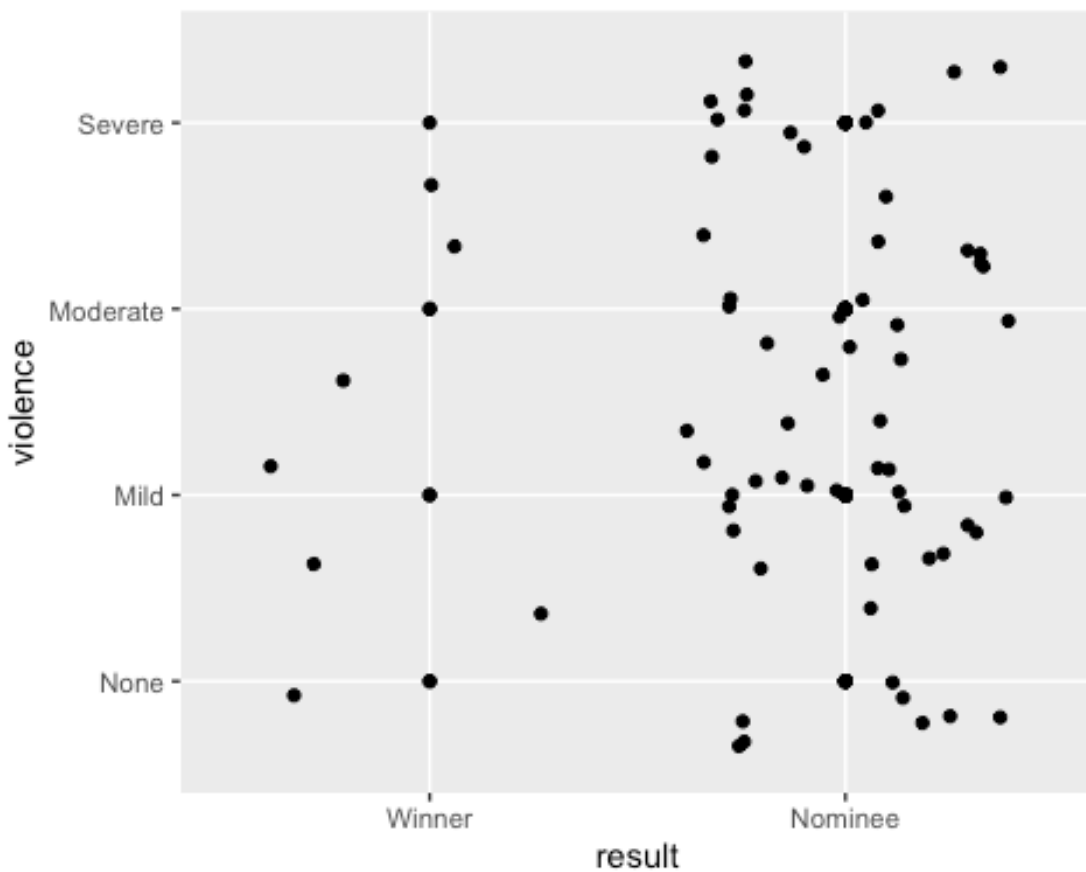
Hypothesis 3

H3 Of Best Picture nominees, winners of the Academy Award for Best Picture are more likely to score a higher level of violence.

Median and Mean H3

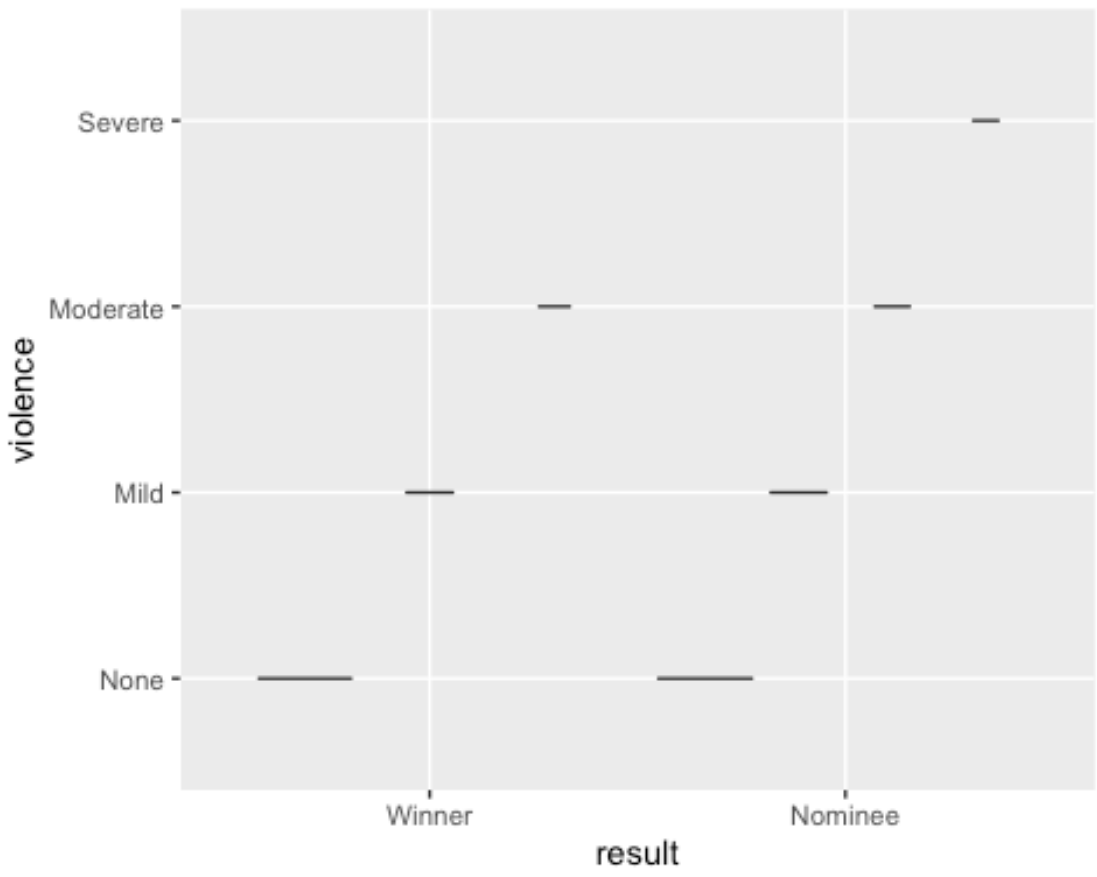
```
## # A tibble: 2 × 5
##   result      n   Mdn     M    SD
##   <ord>   <int> <dbl> <dbl> <dbl>
## 1 Winner      7     2  2.29 1.11
## 2 Nominee    60     2  2.55 0.999
```

Scatter H3

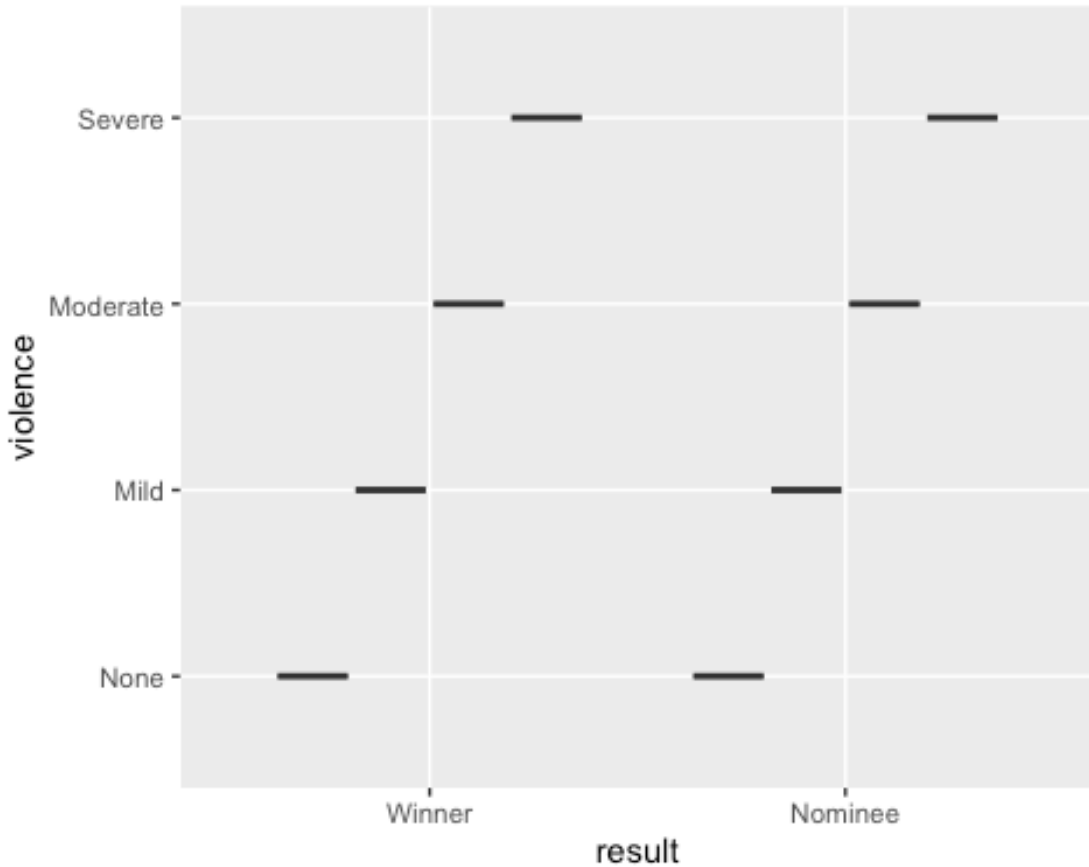


Violin H3

```
## Warning: Groups with fewer than two datapoints have been dropped.
## ℹ Set `drop = FALSE` to consider such groups for position adjustment
purposes.
```


Severe
Moderate
Mild
None
violence
Winner
Nominee
result

Boxplot H3



Spearman Test H3

```
## Warning in cor.test.default(x = mf[[1L]], y = mf[[2L]], ...): Cannot
compute
## exact p-value with ties

##
##  Spearman's rank correlation rho
##
## data:  violence_coded and result_coded
## S = 46230, p-value = 0.5328
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##        rho
## 0.07754618
```

Findings H3

There is no real correlation between violence score and winning the Academy Award Best Picture among Academy Award Best Picture nominees, because p = 0.533. PLEASE CHECK.

Hypothesis 4

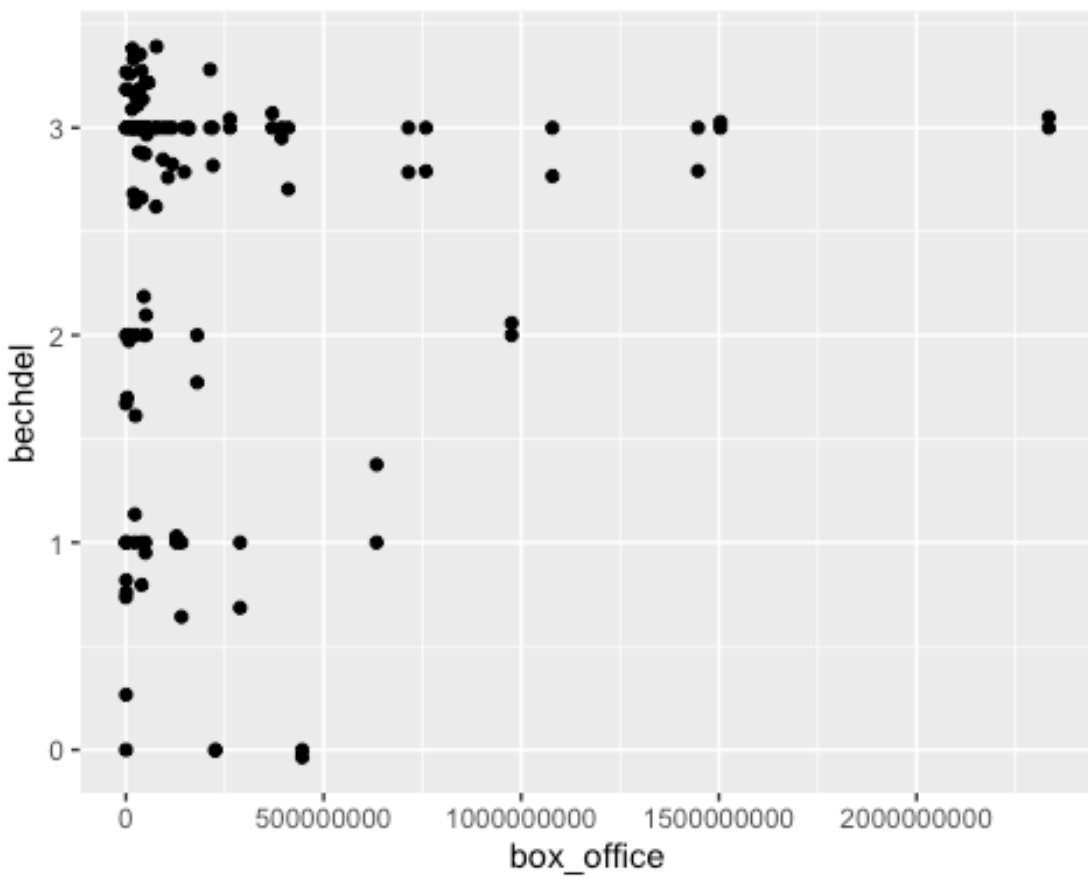
H4 The higher the Box Office revenue for an Academy Award for Best Picture nominated film, the more Bechdel Test criteria the film will pass.

Median and Mean H4

```
## # A tibble: 4 × 5
##   bechdel     n      Mdn          M         SD
##     <dbl> <int>    <dbl>      <dbl>      <dbl>
## 1       0     4       NA         NA         NA
## 2       1    13       NA         NA         NA
## 3       2     8 34831574 160866240  334432862.
## 4       3    42 51471937 265309461. 489957533.
```

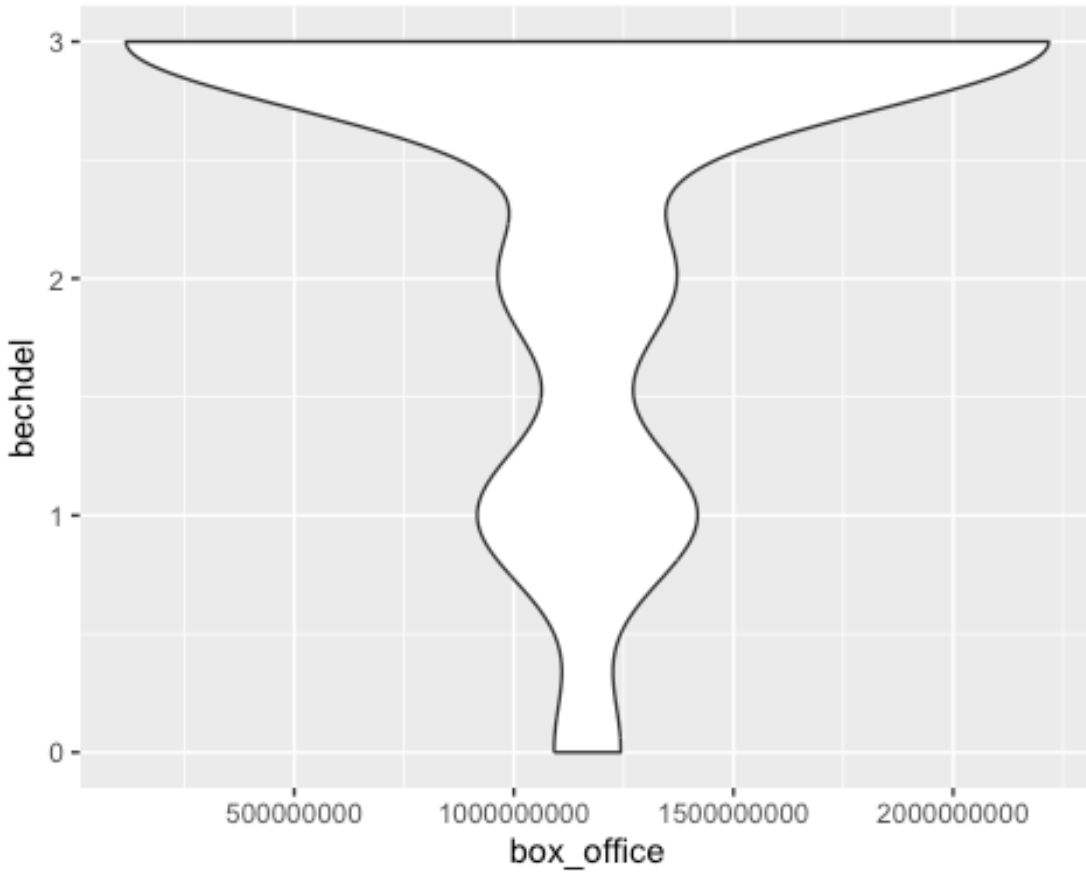
Scatter H4

```
## Warning: Removed 4 rows containing missing values or values outside the
scale range
## (`geom_point()`).
## Removed 4 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



Violin H4

```
## Warning: Removed 4 rows containing non-finite outside the scale range
## (`stat_ydensity()`).
```

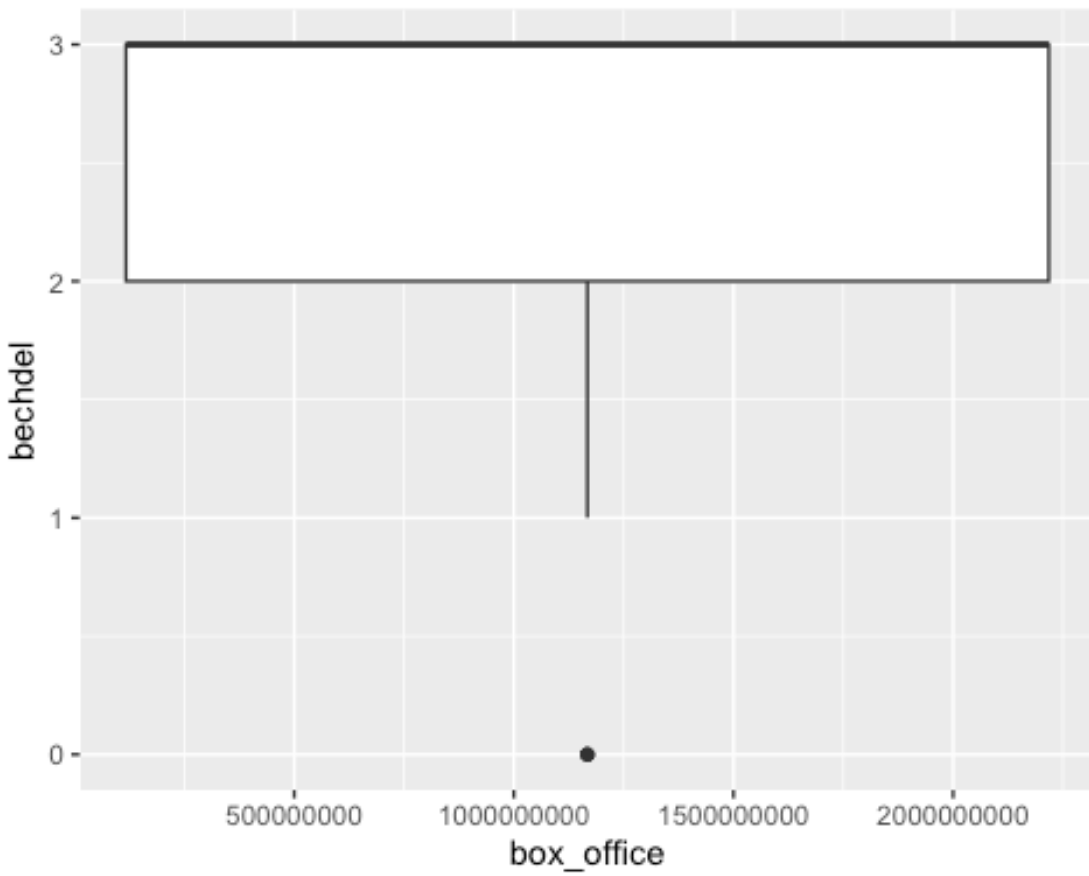


Boxplot H4

```
## Warning: Orientation is not uniquely specified when both the x and y
aesthetics are
## continuous. Picking default orientation 'x'.

## Warning: Continuous x aesthetic
## ℹ did you forget `aes(group = ...)`?

## Warning: Removed 4 rows containing missing values or values outside the
scale range
## (`stat_boxplot()`).
```



Spearman Test H4

```
## Warning in cor.test.default(x = mf[[1L]], y = mf[[2L]], ...): Cannot
compute
## exact p-value with ties

##
##  Spearman's rank correlation rho
##
## data:  bechdel and box_office
## S = 36705, p-value = 0.3529
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.1190165
```

Findings H4

There is no real correlation between Bechdel Test criteria met and Box Office revenue for Academy Award Best Picture nominees, because p = 0.353. PLEASE CHECK.

Hypothesis 5

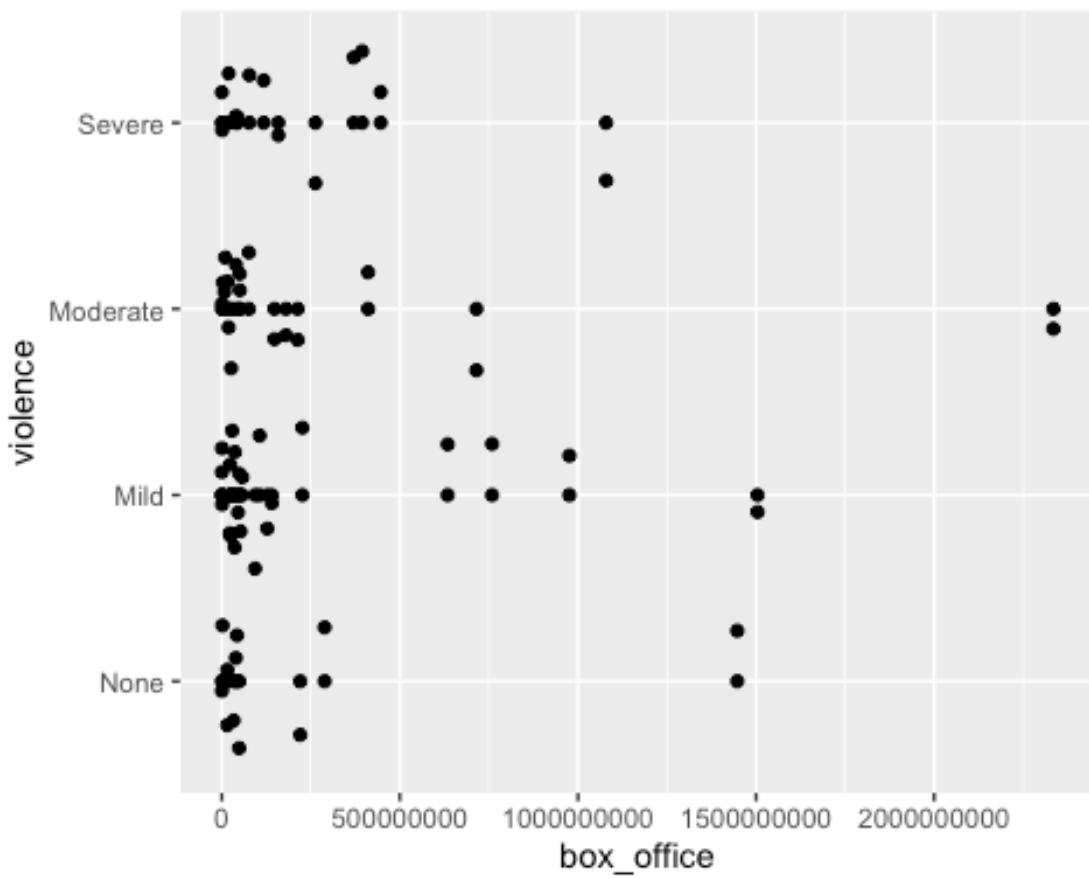
H5 The higher the Box Office revenue for an Academy Award for Best Picture nominated film, the film will score a higher level of violence.

Median and Mean H5

```
## # A tibble: 4 × 5
##   violence     n      Mdn          M         SD
##   <ord>    <int>    <dbl>      <dbl>      <dbl>
## 1 None        11 39458207 195740224. 425661509.
## 2 Mild        24       NA         NA         NA
## 3 Moderate    18       NA         NA         NA
## 4 Severe      14       NA         NA         NA
```

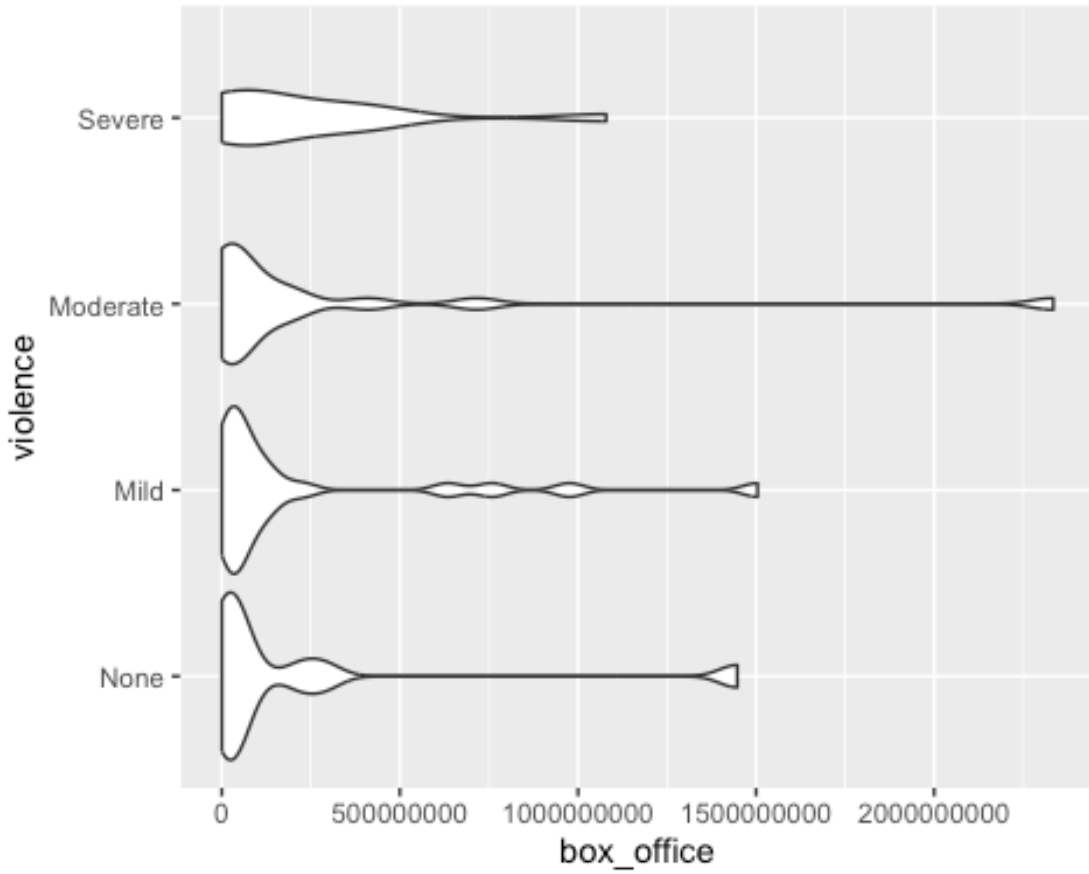
Scatter H5

```
## Warning: Removed 4 rows containing missing values or values outside the
scale range
## (`geom_point()`).
## Removed 4 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



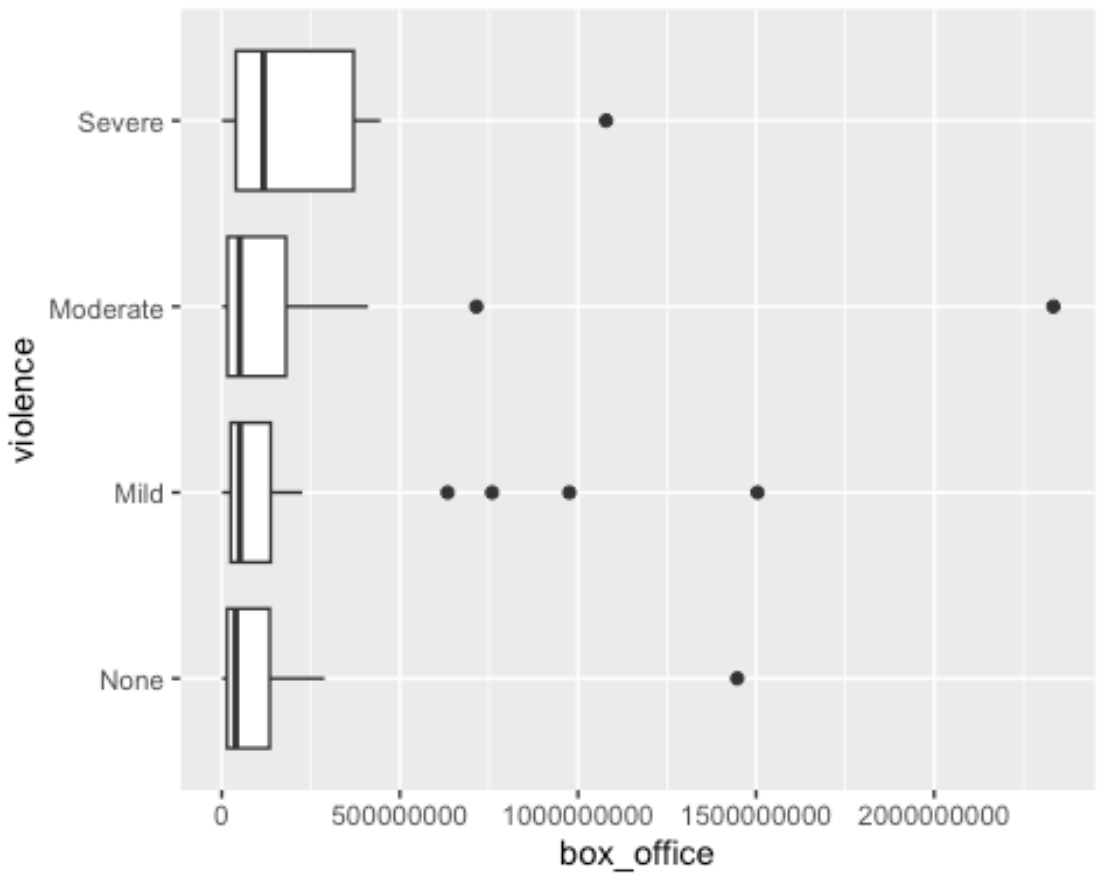
Violin H5

```
## Warning: Removed 4 rows containing non-finite outside the scale range
## (`stat_ydensity()`).
```



Boxplot H5

```
## Warning: Removed 4 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



Spearman Test H5

```
## Warning in cor.test.default(x = mf[[1L]], y = mf[[2L]], ...): Cannot
compute
## exact p-value with ties

##
##  Spearman's rank correlation rho
##
## data:  violence_coded and box_office
## S = 36711, p-value = 0.3534
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.1188849
```

Findings H5

There is no real correlation between violence score and Box Office revenue for Academy Award Best Picture nominees, because p = 0.354. PLEASE CHECK.